

4.4.39. Suppose that in standard factored form

$a = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$, where k is a positive integer: $p_1, p_2, \dots, p_k$ are prime numbers; and $e_1, e_2, \dots, e_k$ are positive integers.

a. What is the standard factored form for a^3 ?

b. Find the least positive integer k such that

$2^4 * 3^5 * 7 * 11^2 * k$ is a perfect cube (that is, it equals an integer to the third power). Write the resulting product as a perfect cube.

a. What is the standard factored form for a^3 ?

$$a = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$$

$$a^3 = (p_1^{e_1} p_2^{e_2} \dots p_k^{e_k})^3 = p_1^{3e_1} p_2^{3e_2} \dots p_k^{3e_k}$$

Solution: $a^3 = p_1^{3e_1} p_2^{3e_2} \dots p_k^{3e_k}$

b. Find the least positive integer k such that

$2^4 * 3^5 * 7 * 11^2 * k$ is a perfect cube (that is, it equals an integer to the third power). Write the resulting product as a perfect cube.

$$a^3 = 2^4 * 3^5 * 7 * 11^2 * k$$

$$2^4 * 2^2 = 2^6$$

$$3^5 * 3^1 = 3^6$$

$$7^1 * 7^2 = 7^3$$

$$11^2 * 11^1 = 11^3$$

$$k = 2^2 * 3^1 * 7^2 * 11^1$$

$$a^3 = 2^4 * 3^5 * 7 * 11^2 * (2^2 * 3^1 * 7^2 * 11^1)$$

$$a^3 = 2^6 * 3^6 * 7^3 * 11^3$$

$$a^3 = (2^2 * 3^2 * 7 * 11)^3$$

Solution: $k = 6468$